

L03 CIRCADIAN LIGHTING DESIGN | O (MAX : 3 PT)

Intent : Support circadian and psychological health through indoor daylight exposure and outdoor views.

Summary : This WELL feature requires projects to provide users with appropriate exposure to light for maintaining circadian health and aligning the circadian rhythm with the day-night cycle.

Issue :

Our body's circadian rhythms are kept in sync by various cues, including light. Humans have evolved to base their circadian rhythms around the natural light-dark patterns associated with daytime and night-time. However, with humans being indoors for extended periods of time, exposure to adequate levels of light have been compromised as typical indoor electric light levels often do not equate to the amount of light the human body traditionally receives outdoors.¹ Light deficiencies affects the functioning of the circadian system and quality of sleep. Disruption of circadian rhythm has been linked with obesity, diabetes, depression and metabolic disorders.²⁻⁴ Exposure to light at night has also been associated with negative health effects, such as breast cancer, circadian phase disruption and sleep disorders.^{5,6}

Solutions : Since circadian response of humans to light is dependent on the light that enters the eye, factors such as spectral properties of the light, brightness levels, duration, and timing of exposure should be considered. The light levels must be achieved on the vertical plane, at the eye level of the occupant to simulate the light entering the eye of the user.^{7,8} It is also important to consider the duration of exposure to light, as well as the timing of exposure. Stimulating the circadian system at night through exposure to bright light can negatively impact sleep quality.^{9,10}

PART 1 MEET LIGHTING FOR DAY-ACTIVE PEOPLE (MAX : 3 PT)

For All Spaces except Dwelling Units & Guest Rooms:

For workstations used during the daytime, electric lighting is used to achieve the following thresholds:

- a. The following light levels are achieved for at least four hours (beginning by noon at the latest) at a height of 45 cm above the work-plane for all workstations in regularly occupied spaces:

Tier	Threshold	Threshold for Projects with Enhanced Daylight	Points
1	At least 150 EML [136 melanopic EDI]	OR At least 120 EML [109 melanopic EDI] and either L05 Part 1 or L06 Part 1	1
2	At least 275 EML [250 melanopic EDI] ¹¹	OR At least 180 EML [163 melanopic EDI] and either L05 Part 1 or L06 Part 1	3

- b. The light levels are achieved on the vertical plane at eye level to simulate the light entering the eye of the occupant.

Note : Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

For Dwelling Units & Guest Rooms:

The following requirements are met in each dwelling unit:

- a.

Electric lighting is used to achieve the following light levels:

Tier	Threshold	Threshold for Projects with Enhanced Daylight	Points
1	At least 150 EML [136 melanopic EDI]	OR At least 120 EML [109 melanopic EDI] and either L05 Part 1 or L06 Part 1	1
2	At least 275 EML [250 melanopic EDI] ¹¹	OR At least 180 EML [136 melanopic EDI] and either L05 Part 1 or L06 Part 1	3

- b. The light levels are dimmable. If automated lighting is used, it is automatically dimmed after 8:00 pm.
- c. The light levels are achieved in living rooms and kitchens at a height of 140 cm in the center of the room. For studio apartments and guestrooms without living rooms, also test in the center of the room. If workstations are present, light levels are achieved at a height of 45 cm above the work-plane.

Note : Refer to the Performance Verification Guidebook for information on sensor/testing requirements, required testing duration and compliance calculations.

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